

Independence of the CH with gentle introduction to Forcing

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- $\{0, 1, 2, 3, \dots, n\} := n + 1$ $n \cup \{n\} := n + 1$

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Definition

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$$0 < 1 < 2 < \dots < \omega < \omega + 1 < \omega + 2 < \dots < \omega + \omega = \omega.2 < \dots < \omega.3 < \dots < \omega.\omega = \omega^2 < \dots < \omega^3 < \dots < \omega^\omega < \dots < \omega^{\omega^\omega} < \omega^{\omega^{\dots}}$$

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$$\mathcal{P}(\omega) = \{x : x \subseteq \omega\} \sim 2^\omega = {}^\omega 2 = \{f : \omega \longrightarrow 2\}$$

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Corollary

- 1 $\mathbb{R}$ is uncountable.

proof: There is no bijection $f : \omega \longrightarrow \mathbb{R}$

- 2 There are infinitely many different uncountable.

$\mathcal{P}(\omega), \mathcal{P}(\mathcal{P}(\omega)), \dots$

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$\aleph_0, \aleph_1, \aleph_3, \dots, \aleph_\omega, \aleph_{\omega+1}, \dots$

$c = \aleph_?$

$f : \omega_? \longrightarrow \mathbb{R}$ is a bijection.

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- 2 Doesn't matter how many axiom you are using, there will always be true statements that cannot be proved.
- 3 If T is consistent, then we can not prove this consistency in T .

consistency of CH and $\neg CH$

Gödel 1939: $Con(ZF)$ implies $Con(ZFC + CH)$

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$$V[G] \models ZFC + c \geq \aleph_2$$

FORCING

Definition

$\langle P, \leq \rangle$ is a partial ordered set if for every $p, q, r \in P$:

- 1 $p \leq p$
- 2 $p \leq q$ and $q \leq p$ implies $p = q$
- 3 $p \leq q$ and $q \leq r$ implies $p \leq r$

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Forcing Notion

Let $V \models ZFC$,

$\mathbb{P} = \langle P, \leq_P \rangle \in V$ a poset with largest element is a forcing notion.

$$Add(\aleph_0, 1) = \{p: \omega \longrightarrow 2 \mid p \text{ is finite}\}$$

$p_1 \leq p_2$ **or** p_1 extends p_2 **or** p_1 is stronger than p_2
or p_1 has more information than p_2 **iff** $p_2 \subseteq p_1$

Dense subsets

$D \subseteq \mathbb{P}$ is dense if

$$\forall p \in \mathbb{P} \exists d \in D (d \leq p)$$

Fix $n \in \omega$, let $D_n = \{p \in \text{Add}(\aleph_0, 1) : n \in \text{dom}(p)\}$

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clearly $q \leq p$ and $q \in D$.

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$G \subseteq \mathbb{P}$ is a filter if

- 1 For all $p, q \in G$, there is $r \in G$ such that $r \leq p$ and $r \leq q$
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Generic Set

G is Generic if

- $G \subseteq \mathbb{P}$ is a filter
- G meets every dense subset of $\mathbb{P}$ that lies in V

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Let V be a (nice) model and ON be the class of all ordinals.

$V \cap ON$ is an ordinal but $V \cap ON \notin V$.

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We can adjoin G to V by an amazing method and get $V[G]$

- 1 $V[G] \models \text{ZFC}$
- 2 $V \subseteq V[G]$ and $G \in V[G]$
- 3 $V[G] \cap ON = V \cap ON$
- 4 If M satisfies (1) – (3), then $V[G] \subseteq M$.

more magics

Theorem

For every formula $\varphi(x_1, \dots, x_n)$ we can find a formula $\varphi'(x_1, \dots, x_n, p, \mathbb{P})$ such that for every forcing notion $\mathbb{P}$ and every $p \in \mathbb{P}$

$V \models \varphi'(x_1, \dots, x_n, p, \mathbb{P})$ iff for every generic $G \subseteq \mathbb{P}$ with $p \in G$

$V[G] \models \varphi(x_1, \dots, x_n)$

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hence there exists n such that $\bigcup G(n) \neq f(n)$



Cohen forcing

$$\text{Add}(\aleph_0, \aleph_2) = \{p : (\omega \times \omega_2) \longrightarrow 2 \mid p \text{ is finite}\}$$

this forcing adds $\aleph_2$ many real.

Cohen showed that $c^{V[G]} \geq \aleph_2^V$

Does $\aleph_2^V = \aleph_2^{V[G]}$?

Thank You